

Shaik Mahammad Khaleel

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SUMMARY

- Hands on experience in synthesis, STA, Floor Planning, Power Planning, Placement, CTS, Routing and IR Drop Analysis.
- Experience in lower technology nodes such as TSMC 16nm and 28nm and able to handle complex designs with multimillion gates and tens of macros.
- Having better understanding of clock tree structure and implementation.
- Hands on experience in low power implementation techniques such as ON-OFF and Multi-Voltage designs.
- Have solid experience in STA, timing Eco's, functional Eco's, metal Eco's implementation.
- Have hands on experience in DRC, LVS, LEC, DRV fixes, Crosstalk Analysis, Glitches, Antenna and IR/EM fixes.
- Proficient in scripting languages TCL, PERL and good at UNIX.

WORK EXPERIENCE

- One year internship as a Physical Design Engineer at RiseTime semiconductors (OPC) Pvt Ltd, from 4th April, 2022 to 22nd April, 2023.

ACADEMIC QUALIFICATIONS

Qualification	Specialization	University/Board	Institute	Year	CGPA/Percentage
B.Tech	EEE	JNTUK	Tirumala Engineering college	2020	8.34 CGPA
Diploma	EEE	AP SBTET	A.M. Reddy Memorial college	2017	85.27%
Matriculation	SSC	Board of Secondary Education Andhra Pradesh	B R C high school	2014	9.2 CGPA

TOOLS

Synthesis	Design Compiler, Genus
Place and Route	ICC2
Timing Analysis	Prime Time
Logical Equivalent Check	Formality

PROJECT DEATILS

BLOCK I

Instance Count	: 2.2M	Number of Macros	: 16
Technology node	: TSMC 28nm	Metal Layers	: 9
Number of Clocks	: 2	Operating Frequency	: 1.2 GHz,
520MHz			
Tools used	: ICC2, PT, Formality.		
Challenges faced	: Congestion issues, local density, clock latencies & skew.		
Role Played	: Floorplanning, Place and Route, DRC clean, STA Signoff.		

BLOCK II

Instance Count	: 2.8M	Number of Macros	: 186
Technology node	: TSMC 16nmX9M	Operating Frequency	: 602MHz
Number of Clocks	: 4		
Challenges faced	: IR fixes		
Tools used	: ICC2, Prime Time, Formality.		
Role Played	: Performed Signoff Timing analysis, Calibre DRC & LVS and Formality checks.		

BLOCK III

Instance Count	: 6.2k	Number of Macros	: 4
Technology node	: TSMC 16nm	Metal Layers	: 9
Number of Clocks	: 1	Operating Frequency	: 150MHz
Challenges faced	: Congestion, fixing max cap, max trans, IR drop analysis DRC and LVS.		
Tools used	: ICC2, Prime Time, Formality.		
Role Played	: Sanity checks, Floorplanning, Placement, CTS, Routing, Timing analysis and Signoff.		

DECLARATION

I do hereby declare that the above particulars of facts and information stated are true, correct, and complete to the best of my belief and knowledge.

Place : Hyderabad

Signature

Date : 23 March, 2023

Shaik Mahammad Khaleel